

Label Studio Annotation Evidence — OBC 2023 §9.8 Gold Set

BIM-Guard rule-extraction evaluation · MAICEN Group 5 · Final Master Project

3 October 2026

1. Purpose

This report documents, with screenshots from the live Label Studio instance, how the human gold set for rule-extraction evaluation was built: the annotation interface, the tasks, what the annotator saw and produced, and how those annotations feed the confusion-matrix scoring of BIM-Guard. All screenshots were captured on 3 October 2026 from the project’s own Label Studio instance (`localhost:8080`, project 1, *BIM-Guard Regulatory Code Annotation*).

Scope. Ontario Building Code 2023, Volume 1, Part 9, Section 9.8 (stairs, ramps, handrails and guards). Dimensional (checkable) rules only.

2. Annotation set-up

Item	Value
Tool	Label Studio (<code>heartexlabs/label-studio</code> , Docker; <code>research/label_studio/docker-compose.yml</code>)
Source text	OBC 2023 Vol. 1 Part 9, converted to DocLang; §9.8 sentences loaded as tasks by the <code>doclang-to-labelstudio</code> skill
Tasks	133 (one per sentence or table), all with a submitted annotation
Pre-annotation	113 tasks carry a machine prediction (<code>bim-guard nlp-v1</code>) that the annotator reviewed and edited
Annotator	One annotator account; no second annotator, no adjudication
Labeling interface	<code>research/label_studio/config.xml</code> — five parts, below

Labeling interface. Each task shows the clause reference and text, then asks for:

- Linguistic spans** (12 labels): deontic modality (`MANDATORY`, `PROHIBITED`, `RECOMMENDED`, `PERMITTED`), conditions (`APPLICABILITY`, `EXCEPTION`, `QUALIFICATION`), dimensions (`DIM_MIN`, `DIM_MAX`, `DIM_EXACT`, `DIM_RANGE`) and `CROSS_REF`.
- IFC target entity class** (e.g. `IfcStairFlight`, `IfcRailing`, `IfcRampFlight`).
- Target property** (e.g. `RiserHeight`, `TreadLength`, `ClearWidth`, `HandrailHeight`).
- Target unit** (`mm`, `m`, `degrees`, `kN`, ...).
- Per-dimension property** (an optional override, so one clause can constrain several properties).

3. Screenshots

3.1 Project overview

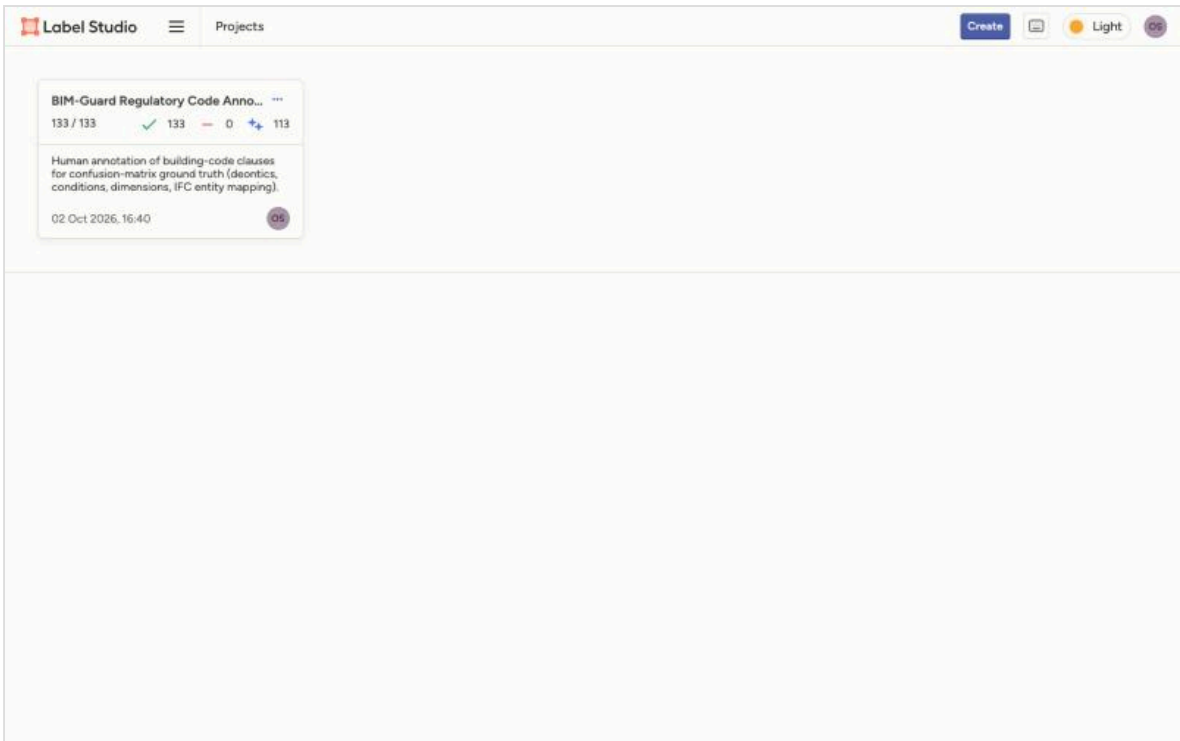


Figure 1. Label Studio project list: 133 of 133 tasks completed, 133 submitted annotations, 113 predictions.

3.2 Data Manager

Label Studio

Projects / BIM-Guard Regulatory Code Annotation

Settings

Light

Default

Actions

Columns

Filters

Order by

Label All Tasks

Import

Export

ID	Completed				Annotated by	section_ref	text
5	Oct 02 2026, 18:59:16	1	0	0	OS	9.8.4.4.(5)	The slope of treads shall not exceed 1 in 50.
6	Oct 02 2026, 18:45:22	1	0	1	OS	9.8.1.1.(1)	This Section applies to design and construction interior and exterior stair
7	Oct 02 2026, 18:45:22	1	0	1	OS	9.8.1.2.(1)	Except as provided in Sentence 9.8.6.2.(3), stairs, ramps, landings, handrai
8	Oct 02 2026, 18:45:22	1	0	1	OS	9.8.1.3.(1)	Where a stair, ramp or landing forms part of an exit , the appropriate
9	Oct 02 2026, 18:45:22	1	0	1	OS	9.8.1.4.(1)	Escalators and moving walks shall conform to the appropriate requirement
10	Oct 02 2026, 18:45:22	1	0	1	OS	9.8.2.1.(1)	Except as provided in Sentence (2) and Article 9.8.4.5A, required exit
11	Oct 02 2026, 18:45:23	1	0	1	OS	9.8.2.1.(2)	Required exit stairs serve a house or an individual

Tasks: 133 / 133 Submitted annotations: 133 Predictions: 113

Figure 2. Data Manager: one row per clause (section_ref, text), with annotation count, annotator and prediction count per task.

3.3 Labeling a clause

Label Studio

Columns

section_ref

1

9.8.4.4.(5)

1

9.8.1.1.(1)

1

9.8.1.2.(1)

1

9.8.1.3.(1)

1

9.8.1.4.(1)

1

9.8.2.1.(1)

1

9.8.2.1.(2)

1

9.8.2.1.(3)

1

9.8.2.1.(4)

1

9.8.2.2.(1)

1

9.8.2.2.(2)

1

9.8.2.2.(3)

1

9.8.3.1.(1)

1

9.8.3.1.(2)

1

9.8.3.1.(3)

1

9.8.3.1.(4)

1

9.8.3.1.(1)

Projects / BIM-Guard Regulatory Code Annotation / Labeling

Settings

Light

OS

Compare All

Osama Ata

bim-guar...nlp-v1

BIM-Guard Regulatory Code Annotation Interface

9.8.4.5A.(1)

Spiral stairs shall have, (a) handrails on both sides, the outer handrail being not less than 1 070 mm high, (b) a clear width not less than 660 mm between handrails, (c) risers that are not more than 240 mm high, (d) treads that, (i) are a minimum of 190 mm deep at a point 300 mm from the centre line of the inside handrail, (ii) have a consistent angle and uniform dimension, and (iii) turn in the same direction, and (e) a clear height not less than 1 980 mm.

1. Linguistic Spans & Operators

MANDATORY 1

PROHIBITED 2

RECOMMENDED 3

PERMITTED 4

APPLICABILITY 5

EXCEPTION 6

QUALIFICATION 7

DIM_MIN 8

DIM_MAX 9

DIM_EXACT 0

DIM_RANGE q

CROSS_REF w

2. IFC Target Entity Class

☒ IfcStairFlight

☐ IfcRailing

☐ IfcRampFlight

☐ IfcDoor

☐ IfcWindow

☐ IfcWall

☐ IfcCovering

☐ IfcSpace

☐ IfcSlab

☐ Other

3. Target Property

☐ Width

☐ RiserHeight

☐ TreadLength

☐ RequiredHeadroom

☐ FlightHeight

☐ NumberOfRiser

☐ HandrailHeight

☒ ClearWidth

☐ FireRating

☐ GuardHeight

☐ GuardLoad

☐ GuardOpeningSize

☐ RampSlope

☐ LandingDimension

☐ HandrailClearance

4. Target Unit

☒ mm

☐ m

☐ degrees

☐ persons

☐ kN

☐ kN/m

☐ kPa

☐ ratio

☐ none

#33

↶

↷

↺

↻

Update

Figure 3. Task #33, OBC 9.8.4.5A.(1) (spiral stairs). The annotator marked “shall” as MANDATORY and six dimension phrases as DIM_MIN / DIM_MAX, set the target to IfcStairFlight, the property to ClearWidth and the unit to mm. A second tab holds the machine pre-annotation for comparison.

Label Studio

Columns

section_ref

1

9.8.4.4.(5)

1

9.8.1.1.(1)

1

9.8.1.2.(1)

1

9.8.1.3.(1)

1

9.8.1.4.(1)

1

9.8.2.1.(1)

1

9.8.2.1.(2)

1

9.8.2.1.(3)

1

9.8.2.1.(4)

1

9.8.2.2.(1)

1

9.8.2.2.(2)

1

9.8.2.2.(3)

1

9.8.3.1.(1)

1

9.8.3.1.(2)

1

9.8.3.1.(3)

1

9.8.3.1.(4)

1

9.8.3.1.(1)

Projects / BIM-Guard Regulatory Code Annotation / Labeling

Settings

Light

OS

Compare All

Osama Ata

bim-guar...nlp-v1

BIM-Guard Regulatory Code Annotation Interface

Table under 9.8.4.1

Table 9.8.4.1.

Rise for Rectangular Treads, Tapered Treads and Winders and Run for Rectangular Treads Forming Part of Sentences 9.8.4.1(1) and 9.8.4.2(1) Stair Type Max. Rise, mm, for All Steps Min. Rise, mm, for All Steps Max. Run , mm, for Rectangular Treads Min. Run , mm, for Rectangular Treads Private stairs (1) 200 125 355 255 Public stairs (2) 180 125 no limit 280 Service stairs (3) no limit 125 355 no limit Stairs to unoccupied attic space (4) no limit 125 355 no limit Stairs to crawl spaces no limit 125 355 no limit Stairs that serve mezzanines not exceeding 20 m 2 within live/work units no limit 125 355 no limit Column 1 2 3 4 5

1. Linguistic Spans & Operators

MANDATORY 1

PROHIBITED 2

RECOMMENDED 3

PERMITTED 4

APPLICABILITY 5

EXCEPTION 6

QUALIFICATION 7

DIM_MIN 8

DIM_MAX 9

DIM_EXACT 0

DIM_RANGE q

CROSS_REF w

2. IFC Target Entity Class

IfcStairFlight

IfcRailing

IfcRampFlight

IfcDoor

IfcWindow

IfcWall

IfcCovering

IfcSpace

IfcSlab

Other

3. Target Property

Width

RiserHeight

TreadLength

RequiredHeadroom

FlightHeight

NumberOfRiser

HandrailHeight

ClearWidth

FireRating

GuardHeight

GuardLoad

GuardOpeningSize

RampSlope

LandingDimension

4. Target Unit

mm

m

degrees

persons

kN

kN/m

kPa

ratio

none

#24

←

→

↺

↻

Update

Figure 4. Task #24, Table 9.8.4.1 (rise and run of treads): the most densely annotated task (42 annotation items). Conditions such as “Private stairs” are APPLICABILITY; each table cell value is a DIM_MIN or DIM_MAX.

3.4 Labeling interface configuration

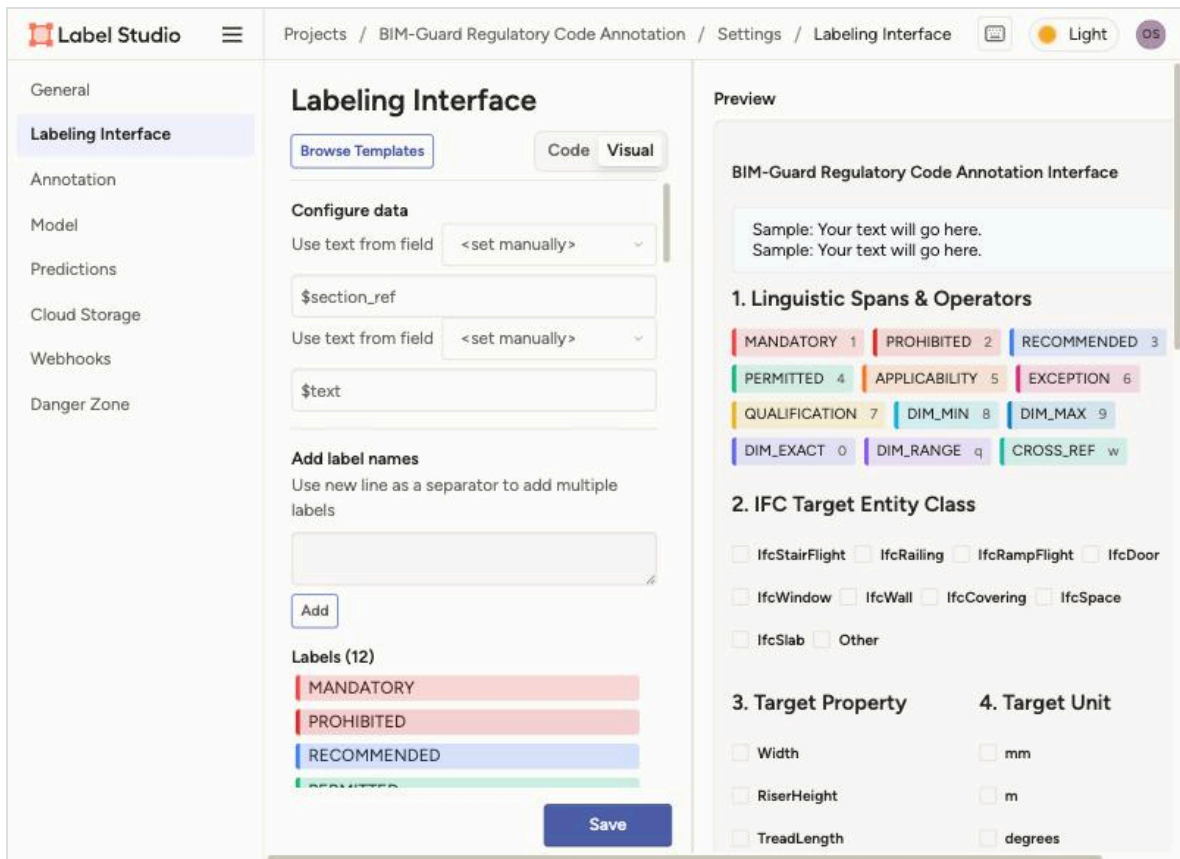


Figure 5. Project Settings → Labeling Interface, showing the `$section_ref` and `$text` data fields, the 12 span labels and the IFC entity, property and unit choices.

4. Annotation statistics

Computed from the Label Studio export `project1_human_2026-10-03.json` (133 tasks, 133 annotations).

Span label	Count	Span label	Count
APPLICABILITY	126	DIM_MAX	40
MANDATORY	102	PERMITTED	14
CROSS_REF	91	QUALIFICATION	8
DIM_MIN	77	PROHIBITED	2
EXCEPTION	49	DIM_RANGE	1
		Total span regions	510

IFC target entity	Annotations
IfcStairFlight	54
IfcRailing	39
IfcSlab	16
IfcRampFlight	7
IfcWindow	5
IfcCovering	4
IfcDoor	2

IFC target entity	Annotations
Other	2
IfcWall	1

5. From annotation to scored evidence

The export is converted by `eval/label_studio_bridge.py` into gold rules (clause reference, IFC target, property, operator, value). The current gold set has **129 live clauses** and **116 gold rules** (four fragment tasks are flagged superseded and excluded).

BIM-Guard extraction was then run through its real web interface (Figure 6) with a Playwright script (`eval/e2e/extraction_confusion_e2e.py`), so the scored drafts are exactly what a reviewer sees, and scored by `eval/score_extraction_vs_human.py`.

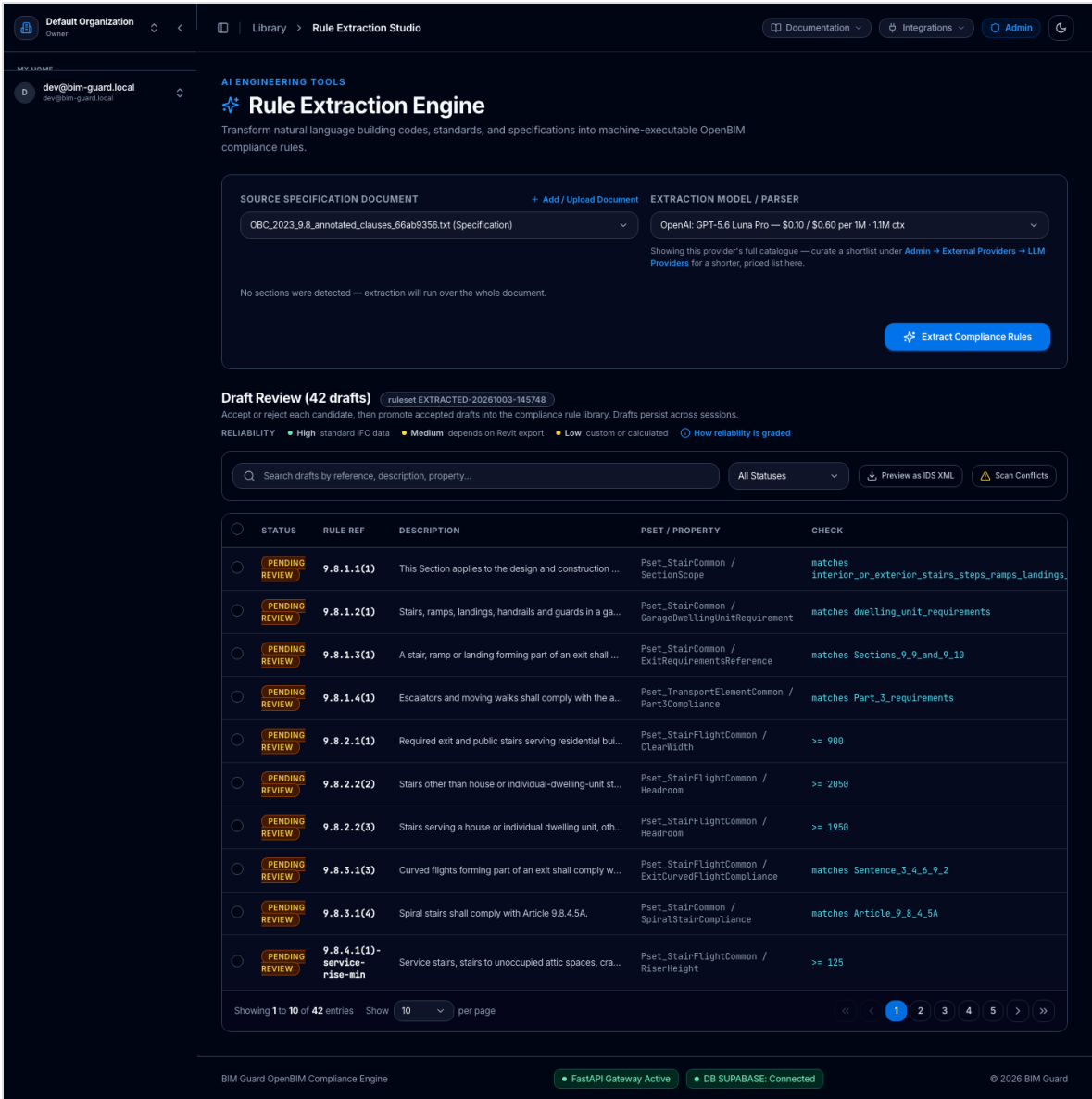


Figure 6. BIM-Guard Rule Extraction Studio, run 4: 42 drafts produced from the annotated clause text, listed for accept/reject review.

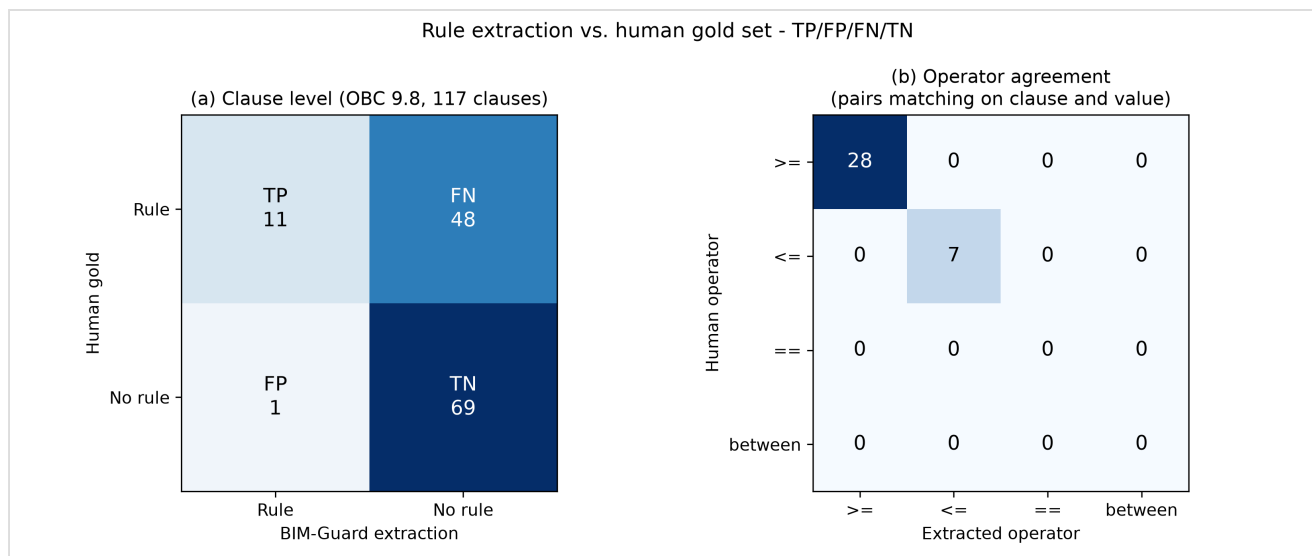


Figure 7. Run 4 confusion matrices against the 3 October gold set (eval/plot_extraction_confusion.py).

Run 4 results (eval/results/e2e/run4_corrected_gold/confusion.md; model openai/gpt-5.6-luna-pro):

Level	TP	FP	FN	TN	Precision	Recall	F1
Clause-level	11	1	48	69	91.7%	18.6%	31.0%
Rule-level, lenient	22	4	94	n/a	84.6%	19.0%	31.0%
Rule-level, normalized	10	18	106	n/a	35.7%	8.6%	13.9%
Rule-level, strict	9	19	107	n/a	32.1%	7.8%	12.5%

Run 4 is one of four runs and must not be quoted alone. Runs 1–3 were scored on an earlier gold revision (89 rules, 117 clauses) with lenient F1 of 77.6%, 34.5% and 27.3%; the two gold versions are not comparable. Runs 2–4 produced drafts for only part of §9.8 because the whole pasted text reaches the model as one chunk, so recall reflects run-to-run coverage more than per-clause accuracy. See eval/results/e2e/README.md and research/CLAIMS.md §3a.

6. Limitations and disclosures

- **Single annotator, no adjudication, no inter-annotator agreement.** The IAA figures produced by eval/score_iaa.py use simulated inputs and are not evidence.
- **Machine pre-annotation.** Tasks were pre-annotated by nlp-v1 and then reviewed and edited by the annotator, as Figure 3 shows.
- **Gold revisions.** The gold set was revised twice (2 and 3 October). Revision 1 followed a config change after run 1; revision 2 fixed sentences the section extractor had dropped, and part of it (the 15 cells of Table 9.8.4.1) was labelled after BIM-Guard was seen extracting them. Both are disclosed in research/CLAIMS.md §3a.
- **Scope.** Dimensional rules only; Table 9.8.7.1 handrail counts remain unlabelled.
- **Not public.** The gold files and clause text are not redistributed, owing to the licensing of the OBC text (docs/DATA_LICENSING.md), so the scoring cannot be reproduced from the repository alone. The screenshots above show short excerpts only.